

COSC2671-Assignment-2 Group 41 Worksheet

Section 1: Team Project Plan

Week	Planned Task	Person Responsible	Status
Week 8	Decide project topic, research direction, dataset focus and team responsibilities	Nathaniel & Bastien	Completed
Week 9	Collect NVIDIA social media dataset and explore dataset structure	Nathaniel	Completed
Week 9	Review assignment requirements and decide analysis workflow	Nathaniel & Bastien	Completed
Week 10	Clean tweet text and prepare processed dataset for analysis	Nathaniel	Completed
Week 10	Extract hashtags, mentions and cashtags for later analysis	Nathaniel	Completed
Week 11	Apply sentiment analysis and create sentiment related outputs	Nathaniel	Completed
Week 11	Begin network analysis using extracted social media entities	Bastien	Completed
Week 12	Compare social media findings with NVDA stock price movement	Bastien	Completed
Week 12	Write and combine report sections	Nathaniel & Bastien	Completed
Week 13	Final check of report, code files, worksheet and submission files	Nathaniel & Bastien	Completed

Section 2: Weekly Timesheets

2.1 Nathaniel Kong — s3964304

Week	Work Completed	Estimated Hours
Week 8	Reviewed the assignment requirements and helped decide the project topic, research direction and dataset focus on NVIDIA/NVDA social media discussion.	2
Week 9	Collected the NVIDIA/NVDA tweet dataset and checked the dataset structure, column names, number of records and initial data quality.	4
Week 9	Conducted initial data exploration, including checking missing values, tweet examples, posting activity and basic dataset patterns.	3
Week 10	Cleaned the tweet text by removing URLs, user mentions, punctuation, numbers, extra spaces and converting text to lowercase.	5
Week 10	Extracted hashtags, mentions and cashtags from the tweet dataset so they could be used for text analysis and later network analysis.	3
Week 11	Applied sentiment analysis to the cleaned tweet text and created sentiment scores/labels for the dataset.	4
Week 11	Analysed common words, hashtags, cashtags and sentiment patterns to support the report findings.	4
Week 12	Prepared processed CSV files for the next stage of the project, including cleaned text, extracted entities and sentiment outputs.	2
Week 12	Wrote report sections related to data collection, data exploration, cleaning, sentiment analysis and text analysis.	5
Week 13	Helped review final report sections, worksheet content and submission files.	2

2.2 Bastien Kara — s4158123

Week	Work Completed	Estimated Hours
Week 8	Reviewed assignment specification, agreed project scope with the team and confirmed NVIDIA / \$NVDA tweets as the dataset focus.	2

Week 8	Set up local development environment, cloned the team repository, installed NetworkX, yfinance and project requirements.	2
Week 9	Drafted the planned network design (nodes, edges, directedness, weighting and filtering decisions) for the hashtag, mention and cashtag co-occurrence graphs.	3
Week 9	Built the hashtag and mention co-occurrence networks, including the helper that converts a list-of-tokens column into a weighted NetworkX graph.	4
Week 10	Built the cashtag co-occurrence network and the \$NVDA ego network, and produced their centrality tables (degree, weighted degree, betweenness and eigenvector).	4
Week 10	Built the directed user-to-mention influence graph and computed weighted in-degree, weighted out-degree and PageRank to interpret user roles.	3
Week 11	Generated the network visualisations and centrality bar charts and saved them to outputs/figures for use in the report.	3
Week 11	Downloaded NVDA daily price data with yfinance, merged it with the daily sentiment summary and produced the price-vs-sentiment and volume-vs-volume overlay plots.	3
Week 11	Computed Pearson and Spearman correlations between sentiment, tweet volume and NVDA returns, and ran the -5 to +5 day lag analysis.	3
Week 12	Debugged a yfinance column-naming issue and resolved a GitHub push failure caused by the full raw dataset being staged; updated .gitignore.	2
Week 12	Wrote report Section 6 (Network Analysis) and Section 7 (Stock Comparison), including figure references and discussion.	4
Week 12	Drafted the joint sections (abstract, discussion, limitations, conclusion and references) for the group to review.	3
Week 13	Final formatting, PDF export, and review of the merged report and worksheet before submission.	2
Total		38

Section 3: Individual Self-Reflection

3.1 Nathaniel Kong — s3964304

For this assignment, my main contribution was the data collection, data exploration, data cleaning, sentiment analysis and text analysis parts of the project. I prepared the NVIDIA/NVDA tweet dataset so it could be used for later stages of analysis. This included checking the dataset structure, cleaning raw tweet text, extracting hashtags, mentions and cashtags, and creating sentiment-related outputs. One challenge was making sure

useful social media and financial terms were not removed during cleaning, especially hashtags and cashtags such as NVDA. I also prepared processed CSV files for the network analysis and stock price comparison stages. Overall, I contributed mainly to the early and middle parts of the project and supported the final report preparation.

3.2 Bastien Kara — s4158123

My contribution to this project was the network analysis and stock-price comparison in Notebook 3. After my teammates cleaned the dataset and produced the daily sentiment summary, I built four networks — hashtag, mention, and cashtag co-occurrence graphs plus a directed user-to-mention network for influence analysis — and compared the daily sentiment signal against NVIDIA's stock price using yfinance, also producing the centrality tables, the \$NVDA ego network, the lag-correlation analysis, and the price-vs-sentiment plots used in Sections 6 and 7. The data side went smoothly because my teammates kept the processed CSVs in a consistent shape, so I could plug straight in; NetworkX was new to me and I spent time learning how to translate co-occurrence into a graph that was actually readable, and filtering to the top-N tokens with a minimum edge weight made a much bigger difference to clarity than I expected. Interpretation was harder than the code — centrality measures are easy to compute but easy to over-claim from, so I revisited the lecture material on degree, betweenness, eigenvector centrality, and PageRank to make sure I was only saying what each measure supports, and the stock comparison was similar in that the correlations are small and the lag analysis is noisy at this sample size, so I wrote the discussion honestly rather than overselling. On the engineering side I hit two problems worth recording: yfinance returned a slightly different column structure on my install, which broke my first attempt at the merge until I pulled the date off the DataFrame index directly, and I accidentally tried to push the full 24 MB raw dataset to GitHub, which the remote rejected, so I cleaned it out of the tracked files and updated .gitignore. Team-wise, splitting the work along the three notebooks and agreeing the processed-CSV schema early worked well; if I did this again I would set up a shared task list from day one — our coordination was ad-hoc and we lost a little time to duplicated work in the first week — and I would scope a finance-specific sentiment model into the project from the start rather than relying on VADER, because that limitation showed up clearly once the results were in.

Section 4: Contribution Summary

Team Member	Main Contributions	Contribution Area
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Nathaniel Kong — s3964304	Data collection, data exploration, data cleaning, hashtag/mention/cashtag extraction, sentiment analysis, text analysis, processed CSV preparation, report writing for early analysis sections	Data preparation and text-based analysis
Bastien Kara — s4158123	Network analysis, network visualisation, stock price comparison, interpretation of later analysis results, report writing for network and stock comparison sections	Network analysis and financial comparison